

# C0 – 2 – ASSESSMENT TOOL 3 – PROBLEM STATEMENT

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COURSE CODE : CSA1704

COURSE NAME : ARTIFICIAL

INTELLIGENCE

## Question 1: Dry Run of A Search Algorithm

Consider the following graph.

(A)

/ \

2 /    \ 4

/        \

(B)---3---(C)

|    \    |

7 |        \ 2 | 3

|        \ |

(D)---2---(E)

\

\ 2

\

(G)

### Heuristic Values

| Node | $h(n)$ |
|------|--------|
|------|--------|

|   |   |
|---|---|
| A | 7 |
|---|---|

|   |   |
|---|---|
| B | 6 |
|---|---|

|   |   |
|---|---|
| C | 4 |
|---|---|

|   |   |
|---|---|
| D | 3 |
|---|---|

|   |   |
|---|---|
| E | 2 |
|---|---|

|   |   |
|---|---|
| G | 0 |
|---|---|

### Task

Perform a dry run of the A\* Search algorithm to find the shortest path from A to G.

For each iteration, show the following:

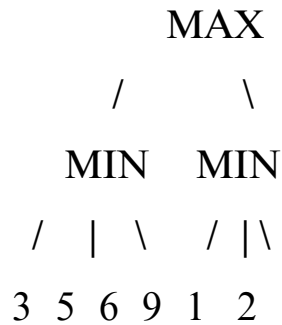
- Current node
- Open List
- Closed List
- $g(n)$
- $h(n)$
- $f(n) = g(n) + h(n)$

### Finally

1. Identify the optimal path.
2. Calculate the total path cost.

## Question 2: Dry Run of Minimax with Alpha-Beta Pruning

Consider the following game tree.



### Task

Perform a dry run of the Minimax algorithm with Alpha-Beta Pruning by evaluating the tree from left to right.

For each node, show:

- Alpha ( $\alpha$ )
- Beta ( $\beta$ )
- Selected value
- Any node(s) that get pruned

### Finally

1. Determine the best move for MAX.
2. State the final minimax value.
3. List the pruned node(s) (if any).